

TASK 3-CYBERSECURITY OWASP X MULEARN BOOTCAMP-VINAYAK D

first of all sorry to delay you guys I consider this as a chance to learn this side by side among many other things i have to do .

STARTED OFF BY DEPLOYING OPENVPN AND ANSWERED SOME QUESTIONS OF TASK 2 AFTER READING THE PASSAGE

Answer the questions below

What networking constructs are used to direct traffic to the right application on a server?

Ports

✓ Correct Answer

How many of these are available on any network-enabled computer?

65535

✓ Correct Answer

[Research] How many of these are considered "well-known"? (These are the "standard" numbers mentioned in the task)

1024

✓ Correct Answer

🔍 Hint

THEN I VENTURED TO TASK 3 AND COMPLETED BY THE BOX THE FOLLOWING

Answer the questions below

What is the first switch listed in the help menu for a "Syn Scan" (more on this later!)?

-sS

✓ Correct Answer

Which switch would you use for a "UDP scan"?

-sU

✓ Correct Answer

If you wanted to detect which operating system the target is running on, which switch would you use?

-O

✓ Correct Answer

Nmap provides a switch to detect the version of the services running on the target. What is this switch?

-sV

✓ Correct Answer

The default output provided by nmap often does not provide enough information for a pentester. How would you increase the verbosity?

-v

✓ Correct Answer

Verbosity level one is good, but verbosity level two is better! How would you set the verbosity level to two?

(Note: it's highly advisable to always use at least this option)

-vv

✓ Correct Answer

We should always save the output of our scans – this means that we only need to run the scan once (reducing network traffic and thus chance of detection), and gives us a reference to use when writing reports for clients.

What switch would you use to save the nmap results in three major formats?

-oA

✓ Correct Answer

What switch would you use to save the nmap results in a "normal" format?

-oN

✓ Correct Answer

A very useful output format: how would you save results in a "grepable" format?

-oG

✓ Correct Answer

Sometimes the results we're getting just aren't enough. If we don't care about how loud we are, we can enable "aggressive" mode. This is a shorthand switch that activates service detection, operating system detection, a traceroute and common script scanning.

How would you activate this setting?

-A

✓ Correct Answer

Nmap offers five levels of "timing" template. These are essentially used to increase the speed your scan runs at. Be careful though: higher speeds are noisier, and can incur errors!

How would you set the timing template to level 5?

-T5

✓ Correct Answer

We can also choose which port(s) to scan.

How would you tell nmap to only scan port 80?

-p 80

✓ Correct Answer

How would you tell nmap to scan ports 1000-1500?

-p 1000-1500

✓ Correct Answer

A very useful option that should not be ignored:

How would you tell nmap to scan all ports?

-p-

✓ Correct Answer

How would you activate a script from the nmap scripting library (lots more on this later!)?

--script

✓ Correct Answer

How would you activate all of the scripts in the "vuln" category?

--script=vuln

✓ Correct Answer

🔍 Hint

TASK 4-----

When port scanning with Nmap, there are three basic scan types. These are:

- TCP Connect Scans (-sT)
- SYN "Half-open" Scans (-sS)
- UDP Scans (-sU)

Additionally there are several less common port scan types, some of which we will also cover (albeit in less detail). These are:

- TCP Null Scans (-sN)
- TCP FIN Scans (-sF)
- TCP Xmas Scans (-sX)

Most of these (with the exception of UDP scans) are used for very similar purposes, however, the way that they work differs between each scan. This means that, whilst one of the first three scans are likely to be your go-to in most situations, it's worth bearing in mind that other scan types exist.

In terms of network scanning, we will also look briefly at ICMP (or "ping") scanning.

Answer the questions below

Read the Scan Types Introduction.



answer needed

✓ Correct Answer

USED THE NMAP COMMANDS ,I HAVE RAN THESE IN MINE AS WELL AS IN OTHER SYSTEMS

TASK 5

Answer the questions below

Which RFC defines the appropriate behaviour for the TCP protocol?

RFC 9293

✓ Correct Answer

🔍 Hint

If a port is closed, which flag should the server send back to indicate this?

RST

✓ Correct Answer

TASK 6

When a packet is sent to an open UDP port, there should be no response. When this happens, Nmap refers to the port as being `open|filtered`. In other words, it suspects it is open, but it could be firewalled. If it gets a `UDP` response (which is very unusual), then the port is marked as `open`. More commonly there is no response, in which case it is sent a second time as a double-check. If there is still no response then the port is marked `open|filtered` and Nmap moves on.

When a packet is sent to a `closed` `UDP` port, the target should respond with an ICMP (ping) packet containing a message that the port is unreachable. This clearly identifies which Nmap marks as such and moves on.

Due to this difficulty in identifying whether a `UDP` port is actually open, `UDP` scans tend to be incredibly slow in comparison to the various TCP scans (in the region of 20 times the first 1000 ports, with a good connection). For this reason it's usually good practice to run an Nmap scan with `--top-ports <number>` enabled. For example, scan the top 20 most commonly used `UDP` ports, resulting in a much more acceptable scan time.

When scanning `UDP` ports, Nmap usually sends completely empty requests -- just raw `UDP` packets. That said, for ports which are usually occupied by well-known services, Nmap sends a protocol-specific payload which is more likely to elicit a response from which a more accurate result can be drawn.

Answer the questions below

If a UDP port doesn't respond to an Nmap scan, what will it be marked as?

open|filtered

✓ Correct

When a UDP port is closed, by convention the target should send back a "port unreachable" message. Which protocol would it use to do so?

ICMP

✓ Correct

task 7

Answer the questions below

If a UDP port doesn't respond to an Nmap scan, what will it be marked as?

open|filtered

✓ Correct Answer

When a UDP port is closed, by convention the target should send back a "port unreachable" message. Which protocol would it use to do so?

ICMP

✓ Correct Answer

task 8

Answer the questions below

Which of the three shown scan types uses the URG flag?

xmas

✓ Correct Answer

Why are NULL, FIN and Xmas scans generally used?

Firewall Evasion

✓ Correct Answer

Which common OS may respond to a NULL, FIN or Xmas scan with a RST for every port?

Microsoft Windows

✓ Correct Answer

task 9----

Answer the questions below

How would you perform a ping sweep on the 172.16.x.x network (Netmask: 255.255.0.0) using Nmap? (CIDR notation)

nmap -sn 172.16.0.0/16

✓ Correct Answer

🔍 Hint

task 10

Answer the questions below

What language are NSE scripts written in?

Lua

✓ Correct Answer

Which category of scripts would be a *very* bad idea to run in a production environment?

intrusive

✓ Correct Answer

task 11

Answer the questions below

What optional argument can the `ftp-anon.nse` script take?

maxlist

✓ Correct Answer

task 12

Answer the questions below

Search for "smb" scripts in the `/usr/share/nmap/scripts/` directory using either of the demonstrated methods. What is the filename of the script which determines the underlying OS of the SMB server?

smb-os-discovery.nse

✓ Correct Answer

Read through this script. What does it depend on?

smb-brute

✓ Correct Answer

🔍 Hint

task 13

Answer the questions below

Which simple (and frequently relied upon) protocol is often blocked, requiring the use of the `-Pn` switch?

ICMP

✓ Correct Answer

[Research] Which Nmap switch allows you to append an arbitrary length of random data to the end of packets?

--data-length

✓ Correct Answer

task 14

Answer the questions below

Does the target ip respond to ICMP echo (ping) requests (Y/N)?

N

✓ Correct Answer

Perform an Xmas scan on the first 999 ports of the target -- how many ports are shown to be open or filtered?

999

✓ Correct Answer

There is a reason given for this -- what is it?

Note: The answer will be in your scan results. Think carefully about which switches to use -- and read the hint before asking for help!

No Response

✓ Correct Answer

🔍 Hint

Perform a TCP SYN scan on the first 5000 ports of the target -- how many ports are shown to be open?

5

✓ Correct Answer

Open Wireshark (see [Cryillic's Wireshark Room](#) for instructions) and perform a TCP Connect scan against port 80 on the target, monitoring the results. Make sure you understand what's going on. Deploy the `ftp-anon` script against the box. Can Nmap login successfully to the FTP server on port 21? (Y/N)

Y

✓ Correct Answer

task 15 conclusion-----

Task 15  Conclusion

You have now completed the Further Nmap room -- hopefully you enjoyed it, and learnt something new!

There are lots of great resources for learning more about Nmap on your own. Front and center are Nmaps own (highly extensive) [docs](#) which have already been mentioned several times throughout the room. These are a superb resource, so, whilst reading through them line-by-line and learning them by rote is entirely unnecessary, it would be highly advisable to use them as a point of reference, should you need it.

Answer the questions below

Read the conclusion.

No answer needed

 Correct Answer



You did it! 🎉 Nmap complete!

Points earned
 328

Completed tasks
 15

Room type
 Walkthrough

Difficulty
 Easy

Streak
 1

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